

LECTURE SHEET

EXERCISE-I

Algebra of Matrices

LEVEL-I (MAIN)

Single answer type questions

1. If $A = (a_{ij})_{2 \times 2}$, where $a_{ij} = i + j$, then A is equal to

1) $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$

2) $\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$

3) $\begin{bmatrix} 1 & 4 \\ 3 & 3 \end{bmatrix}$

4) $\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

2. If $A = \text{diagonal } (3,3,3)$ then $A^4 =$

1) $12A$

2) $81a$

3) $684A$

4) $27A$

3. If $\begin{pmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{pmatrix} \begin{pmatrix} 1 & \tan \theta \\ -\tan \theta & 1 \end{pmatrix} = \begin{pmatrix} a & -b \\ -b & a \end{pmatrix}$

1) $a=1, b=-1$

2) $a=\sec^2 \theta, b=0$

3) $a=0, b=\sin^2 \theta$

4) $a=\sin 2\theta, b=\cos 2\theta$

4. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the correct statement is

1) $A^2 + 5A - 7I = O$

2) $-A^2 + 5A + 7I = O$

3) $A^2 - 5A + 7I = O$

4) $A^2 + 5A + 7I = O$

5. If $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then which one of the following statements is not correct?

1) $A^4 - I = A^2 + I$

2) $A^3 - I = A(A - I)$

3) $A^2 + I = A(A^2 - I)$

4) $A^3 + I = A(A^3 - I)$

6. If $A = \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix}$ and $f(t) = t^2 - 3t + 7$, then $f(A) + \begin{bmatrix} 3 & 6 \\ -12 & -9 \end{bmatrix}$ is equal to

1) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

2) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

3) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

4) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

7. If $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ then $\text{tr}(A) + \text{tr}\left(\frac{ABC}{2}\right) + \text{tr}\left(\frac{A(BC)^2}{4}\right) + \text{tr}\left(\frac{A(BC)^3}{8}\right) + \dots =$

1) 6

2) 9

3) 12

4) 15

8. If $A = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$ and $a^2 + b^2 + c^2 = 1$, then $A^2 =$

1) A

2) 2A

3) 3A

4) 4A

9. If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then

1) $\alpha = a^2 + b^2, \beta = ab$

2) $\alpha = a^2 + b^2, \beta = 2ab$

3) $\alpha = a^2 + b^2, \beta = a^2 - b^2$

4) $\alpha = 2ab, \beta = a^2 + b^2$

10. If $\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} =$

1) $[ax^2 + by^2 + cz^2 + 2hxy + 2gxz + 2fyz]$

2) $[ax^2 + by^2 + cz^2 + hxy + gxz + fyz]$

3) $[2ax^2 + 2by^2 + 2cz^2 + hxy + gxz + fyz]$

4) $[2ax^2 + 2by^2 + 2cz^2 + 2hxy + 2gxz + 2fyz]$

11. If $AB = A$ and $BA = B$ then

1) $A = 2B$

2) $A^2 = A$ and $B^2 = B$

3) $2A = B$

4) cannot be determined

12. If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true?

1) $A = B$

2) $AB = BA$

3) Either A or B is a zero matrix

4) Either A or B is an identity matrix

13. Let $A = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}, B = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, i = \sqrt{-1}$ then

Statement-I : $(A + B)(A - B) = A^2 - B^2$

Statement-II : $AB = BA$

1) Statement 1 is false, statement 2 is true.

2) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

3) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 2.

4) Statement 1 is true, statement 2 is false.

14. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, a, b \in \mathbb{N}$. Then

1) there cannot exist any B such that $AB = BA$

2) there exists more than one but finite number of B 's such that $AB = BA$

3) there exists exactly one B such that $AB = BA$.

4) there exists infinitely many B 's such that $AB = BA$.

15. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then $(aI + bE)^3 =$

1) $aI + bE$

2) $a^3I + b^3E$

3) $a^3I + 3ab^2E$

4) $a^3I + 3a^2bE$

16. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and I is the units matrix of order 2, then A^2 equals

1) $4A - 3I$

2) $3A - 4I$

3) $A - 1$

4) $A + 1$

17. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$ (by the principle of mathematical induction)
- $A^n = nA + (n-1)I$
 - $A^n = 2^{n-1}A + (n-1)I$
 - $A^n = nA - (n-1)I$
 - $A^n = 2^{n-1}A - (n-1)I$
18. If A, B are two square matrices such that $AB=B$; $BA=A$ and $n \in N$ then $(A+B)^n =$
- $2^n(A+B)$
 - $2^{n-1}(A+B)$
 - $2^{n+1}(A+B)$
 - $2^{n/2}(A+B)$
19. If $A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ then the value of $A + A^2 + A^3 + \dots + A^n =$
- A
 - nA
 - $(n+1)A$
 - 0
20. If $A = \begin{bmatrix} \lambda\mu & \mu^2 \\ -\lambda^2 & -\lambda\mu \end{bmatrix}$ then A is
- an idempotent matrix
 - nilpotent matrix
 - an orthogonal matrix
 - symmetric
21. Let A be a 2×2 matrix with non-zero entries and let $A^2 = I$, where I is a 2×2 identity matrix. Define $\text{Tr}(A)$ sum of diagonal elements of A .
- Statement 1 :** $\text{Tr}(A) = 0$
- Statement 2 :** $\text{Tr}(A) \neq 0$
- Statement 1 is false, statement 2 is true.
 - Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.
 - Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 2.
 - Statement 1 is true, statement 2 is false.
22. If $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$, then A is
- idempotent
 - nilpotent
 - involutory
 - symmetric
23. A square matrix A is said to be nilpotent of index m . If $A^m = 0$ now, if for this A , $(I - A)^n = I + A + A^2 + A^3 + \dots + A^{m-1}$, then n is equal to
- 0
 - m
 - $-m$
 - -1
24. If the order of A is 4×3 , the order of B is 4×5 and the order of C is 5×4 , then the order of $A + B + C$ is

26. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to

1) $(2, -1)$ 2) $(-2, 1)$ 3) $(2, 1)$ 4) $(-2, -1)$

27. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $X = P^T Q^{2015} P$, then X is

1) $\begin{bmatrix} 1 & 2015 \\ 0 & 1 \end{bmatrix}$ 2) $\begin{bmatrix} 1 & 0 \\ 2015 & 1 \end{bmatrix}$ 3) $\begin{bmatrix} 2015 & 1 \\ 0 & 1 \end{bmatrix}$ 4) $\begin{bmatrix} 1 & 1 \\ 0 & 2015 \end{bmatrix}$

28. If $\begin{bmatrix} 1 & 3 & 0 \\ 1 & 0 & -2 \\ -4 & -4 & 4 \end{bmatrix} = A + B$ where A is symmetric matrix and B is skew-symmetric then $A - B =$

1) $\begin{bmatrix} 1 & 1 & -4 \\ 3 & 0 & -4 \\ 0 & -2 & 4 \end{bmatrix}$ 2) $\begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 4 \\ 3 & -1 & 2 \end{bmatrix}$ 3) $\begin{bmatrix} 0 & 1 & -1 \\ 2 & 3 & 4 \\ -4 & 1 & 2 \end{bmatrix}$ 4) $\begin{bmatrix} 2 & -3 & 0 \\ 0 & 1 & 2 \\ 2 & 4 & 0 \end{bmatrix}$

29. $\begin{pmatrix} 2 & 3 & 5 \\ 4 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix} = P + Q$, where P is a symmetric and Q is a skew-symmetric then $Q =$

1) $\begin{pmatrix} 0 & -\frac{1}{2} & 2 \\ \frac{1}{2} & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix}$ 2) $\begin{pmatrix} 0 & \frac{1}{2} & 1 \\ -\frac{1}{2} & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$ 3) $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$ 4) $\begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 4 \\ -3 & -4 & 0 \end{pmatrix}$

30. The maximum number of different possible non-zero entries in a skew-symmetric matrix of order ' n ' is

1) $\frac{1}{2}(n^2 - n)$ 2) $\frac{1}{2}(n^2 + n)$ 3) n^2 4) $(n^2 - n)$

31. Let A and B two symmetric matrices of order 3.

Statement 1 : $A(BA)$ and $(AB)A$ are symmetric matrices

Statement 2 : AB is symmetric matrix if matrix multiplication of A with B is commutative.

- 1) Statement 1 is false, statement 2 is true.
- 2) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.
- 3) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.
- 4) Statement 1 is true, statement 2 is false

LEVEL-II (ADVANCED)

Single answer type questions

- If $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$ then $(A-I)(A-2I)(A-3I) =$
 - 1
 - 0
 - A
 - $\frac{1}{2}A$
- If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ then the value of α for which $A^2 = B$ is
 - 1
 - 1
 - 4
 - no real values.
- A and B are two non-singular matrices such that $A^6 = I$ and $AB^2 = BA$ ($B \neq I$). A value of k so that $B^k = I$ is
 - 31
 - 32
 - 64
 - 63
- Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals
 - 52
 - 103
 - 201
 - 205
- If P is a 3×3 matrix such that $P^T = 2P + I$ where P^T is the transpose of P and I is 3×3 identity matrix then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that
 - $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 - $PX = X$
 - $PX = 2X$
 - $PX = -X$

More than one correct answer type questions

- If $A = \begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix}$ (where $\alpha_2 \neq \beta_1$ and $\alpha_1, \alpha_2, \beta_1, \beta_2$ are non-zero) satisfies the equation $x^2 + k = 0$ then
 - $\text{Tra}(A) = 0$
 - $\alpha_1\beta_2 < 0$
 - $\text{Tra}(A) \neq 0$
 - $\alpha_1\beta_2 > 0$
- Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [P_{ij}]$ be a $n \times n$ matrix with $P_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when $n =$
 - 57
 - 55
 - 58
 - 56
- For 3×3 matrices M and N which of the following statement(s) is (are) not correct?
 - $N^T M N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric
 - $MN - NM$ is skew symmetric for all symmetric matrices M and N
 - MN is symmetric for all symmetric matrices M and N
 - $(\text{adj}M)(\text{adj}N) = \text{adj}(MN)$ for all invertible matrices M and N

Linked comprehension type questions**Passage - I :**

Let a, b , and c be three real number satisfying $[a \ b \ c] \begin{bmatrix} 1 & 9 & 7 \\ 8 & 2 & 7 \\ 7 & 3 & 7 \end{bmatrix} = [0 \ 0 \ 0]$

9. If the point $P(a, b, c)$ with reference to (Equation) lies on the plane $2x + y + z = 1$, then the value of $7a + b + c$ is
 a) 0 b) 12 c) 7 d) 6
10. Let ω be a solutions of $x^3 - 1 = 0$ with $\text{Im}(\omega) > 0$. If $a = 2$ with b and c satisfying (Equation), then the value of $\frac{3}{\omega^a} + \frac{1}{\omega^b} + \frac{3}{\omega^c}$ is equal to
 a) -2 b) 2 c) 3 d) -3
11. Let $b = 6$, with a and c satisfying (Equation). If α and β are the roots of the quadratic equation $ax^2 + bx + c$, then $\sum_{n=0}^{\infty} \left(\frac{1}{\alpha} + \frac{1}{\beta} \right)^n$ is
 a) 6 b) 7 c) $\frac{6}{7}$ d) ∞

Passage - II :

If A and B are two square matrices of order 3×3 which satisfy $AB=A$ and $BA=B$, then

12. Which of the following is true ?
 a) If matrix A is singular then matrix B is non-singular
 b) If matrix A is non-singular then matrix B is singular.
 c) If matrix A is singular then matrix B is also singular
 d) Cannot say anything.
13. $(A + B)^7$ is equal to
 a) $7(A + B)$ b) $7 \times I_{3 \times 3}$ c) $64(A + B)$ d) $8I$
14. $(A + I)^5$ is equal to (where I is identity matrix)
 a) $I + 60I$ b) $I + 16A$ c) $I + 31A$ d) $I + 32A$

Matrix matching type questions**15. Column-I****Column-II**

- | | |
|--|---------------------|
| A) $(I-A)^n$ is if A is idempotent | p) $2^{n-1}(I - A)$ |
| B) $(I-A)^n$ is if A is involuntary | q) $I-nA$ |
| C) $(I-A)^n$ is if A is nilpotent of index 2 | r) 0 |
| D) If $A, B, A+I, A+B$ are idempotent matrices, then $AB+BA$ is equal to | s) $I-A$ |

Integer answer type questions

16. If matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$, where a, b, c are real positive numbers, $abc = 1$ and $A^T A = I$, then find the value of $a^3 + b^3 + c^3$.
17. If the product of n matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \dots \dots \dots \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ is equal to the matrix $\begin{pmatrix} 1 & 378 \\ 0 & 1 \end{pmatrix}$ and $n = a^3$ (where $a \in \mathbb{N}$) then $a =$
18. If $a_k = k(10_{C_k}), b_k = (10 - k)(10_{C_k})$ and $A_k = \begin{pmatrix} a_k & 0 \\ 0 & b_k \end{pmatrix}$ and $A = \sum_{k=1}^9 A_k = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ then the sum of digits in $\frac{ab}{2^9 - 1}$ is
19. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, AA^T = 2I$. If $P(\alpha, \beta)$ divides $Q(1, 2)$ and $R(2, 5)$ in the ratio $c : b$ then $\alpha + \beta =$
20. A be the set of 3×3 matrices formed by entries 0, -1 and 1 only. There are three (1), three (-1) and three (0). The number of symmetric matrices with trace $(A) = 0$ is K then $\frac{K}{6}$ is

EXERCISE-II

Determinants

LEVEL-I (MAIN)

Single answer type questions

1. Let the determinant of 3×3 matrix A be 6, there B is a matrix defined by $B = 5A^2$. Then, determinant of B is
 1) 180 2) 100 3) 80 4) 150
2. Matrix A is given by $A = \begin{bmatrix} 6 & 11 \\ 2 & 4 \end{bmatrix}$ then the determinant of $A^{2015} - 6A^{2014}$ is
 1) 2^{2016} 2) $(-11)2^{2015}$ 3) $-2^{2015} \times 7$ 4) $(-9)2^{2014}$
3. If $1, \omega$ and ω^2 are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$ is equal to
 1) 0 2) 1 3) ω 4) ω^2
4. The solutions of the equation $\begin{vmatrix} x & 2 & -1 \\ 2 & 5 & x \\ -1 & 2 & x \end{vmatrix} = 0$ are
 1) 3, -1 2) -3, 1 3) 3, 1 4) -3, -1

5. Let A be a 2×2 matrix with real entries. Let I be the 2×2 identity matrix. Denote by $\text{tr}(A)$, the sum of diagonal entries of A . Assume that $A^2 = I$.

Statement 1 : If $A \neq I$ and $A \neq -I$, then $\det A = -1$.

Statement 2 : If $A \neq I$ and $A \neq -I$, then $\text{tr}(A) \neq 0$.

1) Statement 1 is false, statement 2 is true.

2) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

3) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.

4) Statement 1 is true, statement 2 is false.

6. If $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = k(a-b)(b-c)(c-a)$, then k is equal to

1) -2

2) 1

3) 2

4) abc

7. If the three points $(k, 2k)$, $(2k, 3k)$, $(3, 1)$ are collinear, then k is equal to

1) -2

2) 1

3) $\frac{1}{2}$

4) $-\frac{1}{2}$

8. $\begin{vmatrix} a+b & a & b \\ a & a+c & c \\ b & c & b+c \end{vmatrix}$ is equal to

1) $4abc$

2) abc

3) $a^2b^2c^2$

4) $4a^2bc$

9. The value of the determinant $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$ will be

1) $(a-b-c)(a^2+b^2+c^2)$

2) $(a+b+c)^3$

3) $(a+b+c)(ab+bc+ca)$

4) $(a+b+c)^2$

10. Let a, b and c be such that $b(a+c) \neq 0$. If

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0, \text{ Then the value of } n \text{ is}$$

1) zero

2) any even integer

3) any odd integer

4) any integer

11. If x, y, z are integers in A.P lying between 1 and 9 and $x51, y41$ and $z31$ are three digit numbers,

then the value of $\begin{vmatrix} 5 & 4 & 3 \\ x51 & y41 & z31 \\ x & y & z \end{vmatrix}$ is:

1) $x+y+z$

2) $x-y+z$

3) 0

4) $x+2y+z$

12. If $a_1, a_2, a_3, \dots, a_n, \dots$ are in G.P., then the value of the determinant

$$\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix} \text{ is}$$

- 1) 0 2) -2 3) 2 4) 1

13. If l, m and n are the $p^{\text{th}}, q^{\text{th}}$ and r^{th} terms of a G.P. and are all positive, then $\begin{vmatrix} \log l & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$ equals

- 1) -1 2) 2 3) 1 4) 0

14. If $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0, y \neq 0$, then Δ is

- 1) divisible by neither x nor y 2) divisible by both x and y
3) divisible by x but not y 4) divisible by y but not x

15. If $\Delta_r = \begin{vmatrix} 2^{r-1} & \frac{(r+1)!}{\left(1+\frac{1}{r}\right)} & 4r^3 - 2nr \\ a & b & c \\ 2^n - 1 & (n+1)! - 1 & n^3(n+1) \end{vmatrix}, n \in N$ then $\sum_{r=1}^n \Delta_r =$

- 1) abc 2) $(n+3)!$ 3) 0 4) $a(n!) + b \cdot 2^n + c$

16. If $l_1^2 + m_1^2 + n_1^2 = 1$, etc. and $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$, etc. and $\Delta = \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}$, then $|\Delta| =$

- 1) 4 2) 3 3) 1 4) 2

17. If A and B are different matrices satisfying $A^3 = B^3$ and $A^2 B = B^2 A$, then

- 1) $\det(A^2 + B^2)$ must be zero 2) $\det(A - B)$ must be zero
3) $\det(A^2 + B^2)$ as well as $\det(A - B)$ must be zero.
4) At least one of $\det(A^2 + B^2)$ or $\det(A - B)$ must be zero.

18. If $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = k(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$, then k is equal to

- 1) 1 2) -1 3) $\alpha\beta$ 4) $\alpha\beta\gamma$

19. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$, then $f(x)$ is a polynomial of degree

- 1) 1 2) 0 3) 3 4) 2

20. If α, β, γ are real numbers then $\Delta = \begin{vmatrix} 1 & \cos(\beta-\alpha) & \cos(\gamma-\alpha) \\ \cos(\alpha-\beta) & 1 & \cos(\gamma-\beta) \\ \cos(\alpha-\gamma) & \cos(\beta-\gamma) & 1 \end{vmatrix}$ is equal to

- 1) -1 2) $\cos \alpha \cos \beta \cos \gamma$
3) $\cos \alpha + \cos \beta + \cos \gamma$ 4) 0

21. The number of distinct real roots of the equation, $\begin{vmatrix} \cos x & \sin x & \sin x \\ \sin x & \cos x & \sin x \\ \sin x & \sin x & \cos x \end{vmatrix} = 0$ in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is:

- 1) 4 2) 3 3) 2 4) 1

22. If $C = 2\cos\theta$, then the value of the determinant $\Delta = \begin{vmatrix} C & 1 & 0 \\ 1 & C & 1 \\ 6 & 1 & C \end{vmatrix}$ is

- 1) $\frac{2\sin^2 2\theta}{\sin \theta}$ 2) $8\cos^3 \theta - 4\cos \theta + 6$ 3) $\frac{2\sin 2\theta}{\sin \theta}$ 4) $8\cos^2 \theta + 4\cos \theta + 6$

23. $\begin{vmatrix} 1+\sin^2 \theta & \sin^2 \theta & \sin^2 \theta \\ \cos^2 \theta & 1+\cos^2 \theta & \cos^2 \theta \\ 4\sin 4\theta & 4\sin 4\theta & 1+4\sin 4\theta \end{vmatrix} = 0$, then $\sin 4\theta$ equals to

- 1) $\frac{1}{2}$ 2) 1 3) $-\frac{1}{2}$ 4) -1

24. If $\begin{vmatrix} \operatorname{cosec} \alpha & 1 & 0 \\ 1 & 2\operatorname{cosec} \alpha & 1 \\ 0 & 1 & 2\operatorname{cosec} \alpha \end{vmatrix} = \frac{1}{2} \left(\tan^3 \frac{\alpha}{\lambda} + \cot^3 \frac{\alpha}{\mu} \right)$ then the value of $\lambda + \mu$ is

- 1) 5 2) 2 3) 3 4) 4

25. Maximum value of the expression $\begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix} =$

- 1) 4 2) 6 3) 2 4) -2

26. The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\pi/4 \leq x \leq \pi/4$ is

- 1) 0 2) 2 3) 1 4) 3

27. If $a > 0$ and the discriminant of $ax^2 + 2bx + c$ is negative, then $\begin{vmatrix} a & b & ax+b \\ b & c & bx+c \\ ax+b & bx+c & 0 \end{vmatrix}$ is

- 1) positive 2) equal to $(ac - b^2)(ax^2 + 2bx + c)$
3) negative 4) equal to 0

28. If the value of the determinant $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix}$ is positive then

- 1) $abc < -2$ 2) $abc > 0$ 3) $abc > -8$ 4) $abc = 0$

29. If the entries in a 3×3 determinant are either 0 or 1, then the greatest value of their determinants is

- 1) 1 2) 2 3) 3 4) 9

30. In a third order determinant, each element of the first column consists of sum of three terms, each element of the second column consists sum of four terms, each element of the third column consists of sum of 10 terms, then it can be decomposed into $n!$ determinants then $n =$

- 1) 4 2) 5 3) 6 4) 120

LEVEL-II (ADVANCED)

Single answer type questions

1. If $\omega (\neq 1)$ is a cube root of unity of $A = \begin{bmatrix} 1+2\omega^{100} + \omega^{200} & \omega^2 & 1 \\ 1 & 1+2\omega^{100} + \omega^{200} & \omega \\ \omega & \omega^2 & 2 + \omega^{100} + 2\omega^{200} \end{bmatrix}$, then

- a) A is singular b) $|A| \neq 0$ c) A is symmetric d) none of these

2. If $[.]$ denotes the greatest integer less than or equal to the real number under consideration, and

$-1 \leq x < 0; 0 \leq y < 1; 1 \leq z < 2$, then the value of the determinant $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$ is

- a) $[x]$ b) $[y]$ c) $[z]$ d) $[x]+[y]+[z]$

3. Let A be orthogonal and non-singular matrix of order n , then the determinant of matrix $(A - I_n)$ is equal to

- a) $|I_n - A|$ b) $|A||I_n - A|$ c) $|A|$ d) $(-1)^n |A||I_n - A|$

4. If $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$, where a, b, c are all different, then the

determinant $\begin{vmatrix} 1 & 1 & 1 \\ (x-a)^2 & (x-b)^2 & (x-c)^2 \\ (x-b)(x-c) & (x-c)(x-a) & (x-a)(x-b) \end{vmatrix}$ vanishes when

- a) $a + b + c = 0$ b) $x = \frac{1}{3}(a+b+c)$ c) $x = \frac{1}{2}(a+b+c)$ d) $x = a + b + c$

5. Let M and N be two 3×3 matrices such that $MN=NM$, further, if $M \neq N^2$ and $M^2 = N^4$ then

- a) determinant of $(M^2 + MN^2)$ is 0
b) There is a 3×3 non-zero matrix U such that $(M^2 + MN^2)U$ is zero matrix
c) determinant of $(M^2 + MN^2) \geq 1$
d) for a 3×3 matrix u , if $(M^2 + MN^2)U$ equal to the zero matrix, then U is the zero matrix

6. The value of the determinant of n^{th} order, given by $\begin{vmatrix} x & 1 & 1 & \dots \\ 1 & x & 1 & \dots \\ 1 & 1 & x & \dots \\ \dots & \dots & \dots & \dots \end{vmatrix}$ is

- a) $(x-1)^{n-1}(x+n-1)$ b) $(x-1)^n(x+n-1)$ c) $(x-1)^{-1}(x+n-1)$ d) $(x+1)^{-1}(x-n+1)$

7. The determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$ if

- a) x, y, z are in A.P. b) x, y, z are in G.P. c) x, y, z are in H.P. d) xy, yz, zx are in A.P.

8. $\vec{a}_r = x_r \vec{i} + y_r \vec{j} + z_r \vec{k}, r = 1, 2, 3$ be three mutually perpendicular unit vectors then the value of

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} =$$

- a) zero b) ± 1 c) ± 2 d) ± 3

9. If $\begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & x(x-1)(x+1) \end{vmatrix}$, $f(500)$ is equal to

- a) 0 b) 1 c) 500 d) -500

10. If $a_1 b_1 c_1, a_2 b_2 c_2$, and $a_3 b_3 c_3$ are three - digit even natural numbers and $\Delta = \begin{vmatrix} c_1 & a_1 & b_1 \\ c_2 & a_2 & b_2 \\ c_3 & a_3 & b_3 \end{vmatrix}$, then Δ is

- a) divisible by 2 but not necessarily by 4 b) divisible by 4 but not necessarily by 8
c) divisible by 4 but not necessarily by 6 d) divisible by 6

11. The parameter on which the value of the determinant $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ does not depend on
- a) a b) p c) d d) x

12. In ΔABC , it is being given that

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \cot \frac{A}{2} & \cot \frac{B}{2} & \cot \frac{C}{2} \\ \tan \frac{B}{2} + \tan \frac{C}{2} & \tan \frac{A}{2} + \tan \frac{C}{2} & \tan \frac{A}{2} + \tan \frac{B}{2} \end{vmatrix} = 0$$
 then triangle must be

- a) equilateral b) isosceles c) obtuse angled d) right angled
13. If $a = \cos \theta + i \sin \theta, b = \cos 2\theta - i \sin 2\theta, c = \cos 3\theta + i \sin 3\theta$ and if $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, then
- a) $\theta = 2k\pi, k \in \mathbb{Z}$ b) $\theta = (2k+1)\pi, k \in \mathbb{Z}$ c) $\theta = (4k+1)\pi, k \in \mathbb{Z}$ d) $\theta = (3k+1)\pi, k \in \mathbb{Z}$
14. Let $\{D_1, D_2, D_3, \dots, D_n\}$ be the set of third - order determinants that can be made with the distinct non - zero real numbers $a_1, a_2, \dots, a_9$. Then

- a) $\sum_{i=1}^n D_i = 1$ b) $\sum_{i=1}^n D_i = 0$ c) $D_i = D_j, \forall i, j$ d) $D_i = D_j, \forall i, j$

More than one correct answer type questions

15. If $\Delta = \begin{vmatrix} -x & a & b \\ b & -x & a \\ a & b & -x \end{vmatrix}$, then a factor of Δ is

- a) $a + b + x$ b) $x^2 - (a-b)x + a^2 + b^2 + ab$
c) $x^2 + (a+b)x + a^2 + b^2 - ab$ d) $a + b - x$

16. Which of the following values of α satisfy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$?
- a) -4 b) 9 c) -9 d) 4

17. If $\begin{vmatrix} yz - x^2 & zx - y^2 & xy - z^2 \\ xz - y^2 & xy - z^2 & yz - x^2 \\ xy - z^2 & yz - x^2 & zx - y^2 \end{vmatrix} = \begin{vmatrix} r^2 & u^2 & u^2 \\ u^2 & r^2 & u^2 \\ u^2 & u^2 & r^2 \end{vmatrix}$, then

- a) $r^2 = x + y + z$ b) $r^2 = x^2 + y^2 + z^2$ c) $u^2 = yz + zx + xy$ d) $u^2 = xyz$

18. If $\phi(\alpha, \beta) = \begin{vmatrix} \cos \alpha & -\sin \alpha & 1 \\ \sin \alpha & \cos \alpha & 1 \\ \cos(\alpha + \beta) & -\sin(\alpha + \beta) & 1 \end{vmatrix}$, then

a) $f(300, 200) = f(400, 200)$

b) $f(200, 400) = f(200, 600)$

c) $f(100, 200) = f(200, 200)$

d) $f(100, 300) = f(150, 350)$

19. If maximum and minimum values of $\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$ are α, β then which of the following is/are true?

a) $\alpha + \beta^{99} = 4$

b) $\alpha^3 - \beta^{17} = 26$

c) $\alpha^{2n} - \beta^{2n}$ is always an even integer for $n \in \mathbb{N}$

d) A triangle can be constructed having its sides as $\alpha, \beta, \alpha - \beta$

20. If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2, (x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2, (x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$

and $4 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \lambda \{ \lambda^3 - (\lambda_1 + \lambda_2 + \lambda_3) \lambda^2 + (\lambda_1 \lambda_2 + \lambda_2 \lambda_3 + \lambda_3 \lambda_1) \lambda - \lambda_1 \lambda_2 \lambda_3 \}$ then

a) $\lambda \geq \frac{3}{2}(\lambda_1 \lambda_2 \lambda_3)^{\frac{1}{3}}$

b) $\lambda_1 \lambda_2 \lambda_3 = 8abc$

c) $\sum \lambda_1 \lambda_2 = 4 \sum ab$

d) $2\lambda = \lambda_1 + \lambda_2 + \lambda_3$

Linked comprehension type questions

Passage - I :

Let α, β be the roots of $ax^2 + bx + c = 0$ and let $S_n = \alpha^n + \beta^n$, for $n \geq 1$ and $\Delta = \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$

21. If $\Delta_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}$ then $\frac{\Delta}{\Delta_1^2}$ is equal to

a) 1

b) 2

c) 0

d) 3

22. Value of Δ in terms of α, β is

a) $((\alpha+1)(\alpha-\beta)(\beta-1))^2$

b) $((\alpha-1)(\alpha-\beta)(\beta+1))^2$

c) $((\alpha-1)(\alpha+\beta)(\beta-1))^2$

d) $((\alpha-1)(\alpha-\beta)(\beta-1))^2$

23. Value of Δ in terms of a, b, c is

a) $\frac{(a+b+c)^2(b^2-4ac)}{a^4}$

b) $\frac{(a+b+c)^2(b^2+4ac)}{a^4}$

c) $\frac{(a+b-c)^2(b^2+4ac)}{a^4}$

d) $\frac{(a+b+c)^2(b^2-4ac)}{b^4}$

Passage - II :

Let $\Delta = \begin{vmatrix} -bc & b^2 + bc & c^2 + bc \\ a^2 + ac & -ac & c^2 + ac \\ a^2 + ab & b^2 + ab & -ab \end{vmatrix}$ and the equation $px^3 + qx^2 + rx + s = 0$ has roots a, b, c where

$a, b, c \in \mathbb{R}^+$.

24. The value of Δ is

- a) r^3 / p^3 b) r^2 / p^2 c) $-s/p$ d) $\frac{r^2}{p^3}$

25. The value of Δ is

- a) $\leq 9r^2 / p^2$ b) $\geq 27s^2 / p^2$ c) $\leq 27s^3 / p^3$ d) $\geq \frac{27s}{p}$

26. If $\Delta = 27$ and $a^2 + b^2 + c^2 = 3$, then

- a) $3p + 2q = 0$ b) $4p + 3q = 0$ c) $3p + q = 0$ d) $3p + 3q = 0$

Passage - III :

Consider the polynomial function $f(x) = \begin{vmatrix} (1+x)^a & (1+2x)^b & 1 \\ 1 & (1+x)^a & (1+2x)^b \\ (1+2x)^b & 1 & (1+x)^a \end{vmatrix}$ and a, b being positive integers.

27. The constant term in $f(x)$ is

- a) 2 b) 1 c) -1 d) 0

28. The coefficient of x in $f(x)$ is

- a) 2^a b) $2^a - 3 \times 2^b = 1$ c) 0 d) 2^b

29. Which of the following is true?

- a) All root of the equation $f(x) = 0$ are positive b) All root of the equation $f(x) = 0$ are negative

Matrix matching type questions

33. The entries in a 3×3 determinant are either 1 or -1, then

Column - I

- A) Total number of such determinants
 B) The sum of maximum & minimum trace of the determinants possible
 C) The maximum value of such a determinant
 D) The maximum value of trace of such determinant

Column - II

- p) 4
 q) 3
 r) 512
 s) zero

Integer answer type questions

34. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is

35. Given $(1+x+x^2)^8 = \sum_{k=0}^{16} a_k x^k$ then the value of $\begin{vmatrix} a_3 & a_7 & a_{13} \\ a_4 & a_6 & a_{12} \\ a_5 & a_4 & a_{11} \end{vmatrix}$ is

36. If $a_1, a_2, a_3, 5, 4, a_7, a_8, a_9$ are in H.P. and $D = \begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ then the value of $[D]$ is (where $[]$ represents G.I.F)

37. If $f(x)$ satisfies the equation $\begin{vmatrix} f(x-2) & f(x+5) & f(x-2) \\ 5 & 4 & -5 \\ 10 & 12 & 30 \end{vmatrix} = 0$ for all real x then $f(x)$ is periodic with period

38. If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$, then the value of $(\Delta_{\max})/2$ is _____.

39. If $D_i = \begin{vmatrix} 1 & n & n \\ 2i & n^2+n+1 & n^2+n \\ 2i-1 & n^2 & n^2+n+1 \end{vmatrix}$ and $\sum_{i=1}^n D_i = 56$ then n must be

40. If $D = \begin{vmatrix} 10! & 11! & 12! \\ 11! & 12! & 13! \\ 12! & 13! & 14! \end{vmatrix}$ then $k/3$, where k is the total number of divisor of $\frac{D}{(10!)^3} - 4$, is

41. If $\Delta(x) = \begin{vmatrix} 0 & 2x-2 & 2x+8 \\ x-1 & 4 & x^2+7 \\ 0 & 0 & x+4 \end{vmatrix}$ and $f(x) = \sum_{j=1}^3 \sum_{i=1}^3 a_{ij} c_{ij}$, where a_{ij} is the element of i^{th} row and j^{th} column in $\Delta(x)$ and c_{ij} is the cofactor of a_{ij} , for all i, j then greatest value of $f(x)$, for $x \in [-3, 18]$ is

42. Let k be a positive real number and let $A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$ then $[k]$ is equal to [Note : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k]

43. Given $A = \begin{bmatrix} a & b & 2c \\ d & e & 2f \\ l & m & 2n \end{bmatrix}, B = \begin{bmatrix} f & 2d & e \\ 2n & 4l & 2m \\ c & 2a & b \end{bmatrix}$, then the value of B/A is _____.

EXERCISE-III

Inverse of a matrix

LEVEL-I (MAIN)

Single answer type questions

- If $A = \begin{bmatrix} a & c & b \\ b & a & c \\ c & b & a \end{bmatrix}$ then the cofactor of a_{32} in $A + A^T$ is
 1) $-(2a(b+c) - (b+c)^2)$ 2) $ac - b^2$ 3) $a^2 - bc$ 4) $2a(a+c) - (a+c)^2$
- If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = AA^T$, then $5a+b$ is equal to
 1) 5 2) 4 3) 13 4) -1
- The inverse of the matrix $\begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$ is
 1) $\frac{1}{11} \begin{bmatrix} 1 & 2 \\ -3 & 5 \end{bmatrix}$ 2) $\begin{bmatrix} 1 & 2 \\ -3 & 5 \end{bmatrix}$ 3) $\frac{1}{13} \begin{bmatrix} -2 & 5 \\ 1 & 3 \end{bmatrix}$ 4) $\begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$
- If $A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 8 \end{bmatrix}$, then $A^{-1} =$
 1) $\begin{bmatrix} \frac{1}{9} & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$ 2) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & 8 \end{bmatrix}$ 3) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$ 4) $\begin{bmatrix} \frac{1}{9} & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$
- If $\begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}^{-1}$ then $\alpha =$
 1) 0 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$

MATRICES SYNOPOSIS & LS

(Objective)

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OBJECTIVE MATHEMATICS IA PART-1

MATRICES

6. The inverse of $\begin{bmatrix} 1 & a & b \\ 0 & x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is $\begin{bmatrix} 1 & -a & -b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $x =$
- 1) a 2) b 3) 0 4) 1
7. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$ then
- 1) $a=2, c=1/2$ 2) $a=1, c=-1$ 3) $a=-1, c=1$ 4) $a=1/2, c=1/2$
8. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of A , then α is
- 1) -2 2) 5 3) 2 4) -11
9. If $S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, $A = \frac{1}{2} \begin{bmatrix} b+c & c-a & b-a \\ c-b & c+a & a-b \\ b-c & a-c & a+b \end{bmatrix}$, then $SAS^{-1} =$
- 1) $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ 2) $\frac{1}{2} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ 3) $2 \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ 4) $3 \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$
10. Let P and Q be 3×3 matrices $P \neq Q$. If $P^3 = Q^3$ and $P^2Q = Q^2P$, then determinant of (P^2+Q^2) is equal to
- 1) -2 2) 1 3) 0 4) -1
11. Let $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$. The only correct statement about the matrix A is
- 1) A is a zero matrix 2) $A^2 = I$.
3) A^{-1} does not exist 4) $A = (-1)I$, where I is a unit matrix
12. If $|A| \neq 0$ and $(A-2I)(A-3I) = 0$ then $A^{-1} =$
- 1) $\frac{A-5I}{6}$ 2) $\frac{5I-A}{5}$ 3) $\frac{5A-I}{6}$ 4) $\frac{5I-A}{6}$
13. If $A^2 - A + I = 0$, then the inverse of A is
- 1) $A+I$ 2) A 3) $A-I$ 4) $I-A$

15. Let $A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

I) $A_\alpha A_\beta = A_{\alpha+\beta}$

II) $A_\alpha B_\beta = A_{\alpha\beta}$

III) $(A_\alpha)^{-1} = -A_\alpha$

IV) $(A_\alpha)^{-1} = A_{-\alpha}$

Then which of the above statements is / are correct

1) only II and III

2) only II and IV

3) only I and III

4) only I and IV

16. If $P = \begin{bmatrix} 1 & 11 & 3 \\ 1 & 3 & 3 \\ \alpha & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to

1) 2

2) 5

3) 0

4) 4

17. If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $(\text{Adj} A)^{-1} =$

1) I

2) A

3) $-A$

4) 0

18. Statement-I: There are only finitely many 2×2 matrices which commute with the matrix $\begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}$

Statement-II: If A is non-singular, then it commutes with I , $\text{adj} A$ and A^{-1}

1) Statement 1 is false, statement 2 is true.

2) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.

3) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 2.

4) Statement 1 is true, statement 2 is false.

19. If A is a matrix of order 2×2

Statement-I : $\text{adj}(\text{adj} A) = A$

Statement-II : $|\text{adj} A| = |A|$.

1) only I true

2) only II true

3) I and II true

4) I and II false

20. If $A = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix}$, then $[A(\text{adj} A)A^{-1}]A =$

1) $2 \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

2) $\begin{bmatrix} -6 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & -6 \end{bmatrix}$

3) $\begin{bmatrix} 0 & \frac{1}{6} & -\frac{1}{6} \\ \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \end{bmatrix}$

4) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

21. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \end{bmatrix}$ is the adjoint of a matrix A of order 2×2 and $|A| = 1$, then $\alpha =$

22. Matrix $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if $xyz = 60$ and $8x + 4y + 3z = 20$, then $A(\text{adj}A)$ is equal to

1) $\begin{bmatrix} 64 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 64 \end{bmatrix}$

2) $\begin{bmatrix} 88 & 0 & 0 \\ 0 & 88 & 0 \\ 0 & 0 & 88 \end{bmatrix}$

3) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$

4) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$

23. If A, B and C are $n \times n$ matrices and $\det(A) = 2, \det(B) = 3$ and $\det(C) = 5$, then the value of the $\det(A^2BC^{-1})$ is equal to

1) $\frac{6}{5}$

2) $\frac{12}{5}$

3) $\frac{18}{5}$

4) $\frac{24}{5}$

24. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ if u_1 and u_2 are column matrices such that $Au_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $Au_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ then $u_1 + u_2$ equal to

1) $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

2) $\begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$

3) $\begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$

4) $\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

25. If A, B, C are three square matrices of same order satisfying $A^2 = A^{-1}$ and let $B = A^{2^n}$ and $C = A^{2^{n-2}}$ ($n \in N$) then $\det(B-C) =$

1) 0

2) 1

3) -1

4) $(-1)^n$

26. If $A = [a_{ij}]_{i,j=1}^7$ such that $a_{ij} = \begin{cases} 2 & \text{when } i = j \\ 0 & \text{when } i \neq j \end{cases}$ then $\left\{ \frac{\det(\text{adj}(\text{adj}A))}{7} \right\}$ is (where $\{ \}$ represents fractional part of x) is

1) $\frac{1}{7}$

2) $\frac{2}{7}$

3) $\frac{3}{7}$

4) $\frac{9}{7}$

27. If A and B are square matrices of same order and A is non-singular, then for a positive integer ' n ', $(A^{-1}BA)^n$ is equal to

1) $A^{-n}B^nA^n$

2) $A^nB^nA^{-n}$

3) $A^{-1}B^nA$

4) $n(A^{-1}BA)$

28. If A is an 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1}A'$, then BB' equals

1) $I+B$

2) I

3) B^{-1}

4) $(B^{-1})'$

29. The number of 3×3 non-singular matrices with four entries as 1 and all other entries as 0 is

1) at least 7

2) less than 4

3) 5

4) 6

LEVEL-II (ADVANCED)

Single answer type questions

- Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$, where each of a, b and c is either ω or ω^2 . Then the number of distinct matrices in the set S is
 a) 2 b) 6 c) 4 d) 8
- Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A^{-1} = \left[\frac{1}{6}(A^2 + cA + dI) \right]$. The value of c and d are
 a) $(-6, -11)$ b) $(6, 11)$ c) $(-6, 11)$ d) $(6, -11)$

More than One correct answer Type Questions

- If $\begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta \\ -\tan \theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ then
 a) $a = \cos 2\theta$ b) $a = 1$ c) $b = \sin 2\theta$ d) $b = 1$
- Let $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, $B = \begin{bmatrix} \cos 2\beta & \sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{bmatrix}$ where $0 < \beta < \frac{\pi}{2}$ then which of the following is(are) true
 a) $(AB)^2 = I$ b) $(AB)^{-1} = AB$ c) $BAB = A^{-1}$
 d) The least positive value of α for which $BA^4B = A^{-1}$ is $\frac{2\pi}{3}$
- Which of the following statements are false
 a) If A and B are square matrices of same order such that $ABAB=0$ then it follows that $BABA = 0$
 b) Let A and B be different $n \times n$ matrices with real entries. If $A^3 = B^3$ and $A^2B = B^2A$ then $A^2 + B^2$ is invertible
 c) If A is square singular and symmetric matrix then $\left((A^{-1})^{-1} \right)^{-1}$ is skew symmetric
 d) The matrix of product of two invertible square matrices of same order is also invertible.
- Which of the following statements is/are true about square matrix A of order n ?
 a) $(-A)^{-1}$ is equal to $-A^{-1}$ when n is odd only.
 b) If $A^n = O$, then $I + A + A^2 + \dots + A^{n-1} = (I - A)^{-1}$.
 c) If A is skew-symmetric matrix of odd order, then its inverse does not exist.
 d) $(A^{-1})^T = (A^T)^{-1}$ holds always.

7. P is a non-singular matrix and A, B are two matrices such that $B = P^{-1}AP$ then the true statements among the following are
- a) A is invertible iff B is invertible
 b) $B^n = P^{-1}A^nP \forall n \in \mathbb{N}$
 c) $\forall \lambda \in \mathbb{R}, B - \lambda I = P^{-1}(A - \lambda I)P$
 d) A, B are both singular matrices
8. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$ then the possible values of the determinant of P is (are)
- a) -2
 b) -1
 c) 1
 d) 2
9. If a, b, c are the roots of $x^3 + x^2 - 4 = 0$ and $A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ then
- a) A is non singular
 b) A is orthogonal
 c) $AA^T = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$
 d) $A^{-1} = A^T$
10. If A is an invertible matrix, then $(\text{adj } A)^{-1}$ is equal to
- a) $\text{adj } (A^{-1})$
 b) $\frac{A}{\det A}$
 c) A
 d) $(\det A) A$

Linked comprehension type questions

Paragraph - I :

Let A be a $m \times n$ matrix, if there exists a matrix ' L ' of type $n \times m$ such that $LA = I_n$, then ' L ' is called left inverse of A . Similarly if there exists a matrix ' R ' of type $n \times m$ such that $AR = I_m$ then ' R ' is called right inverse of A .

11. Which of the following is not left inverse of $\begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{pmatrix}$
- a) $\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -1 & 1 & 0 \\ \frac{-1}{2} & \frac{1}{2} & 0 \end{pmatrix}$
 b) $\begin{pmatrix} 2 & -7 & 0 \\ -1 & 1 & 0 \\ \frac{-1}{2} & \frac{1}{2} & 0 \end{pmatrix}$
 c) $\begin{pmatrix} -1 & \frac{1}{2} & 0 \\ \frac{-1}{2} & \frac{1}{2} & 0 \\ \frac{-1}{2} & \frac{1}{2} & 0 \end{pmatrix}$
 d) $\begin{pmatrix} 0 & 3 & -1 \\ -1 & 1 & 0 \\ \frac{-1}{2} & \frac{1}{2} & 0 \end{pmatrix}$
12. Number of right inverse matrix of the matrix $\begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 2 & 3 \end{pmatrix}$ is
- a) 0
 b) 1
 c) 2
 d) infinitely many
13. For which of the following matrices number of left inverses is greater than the number of right inverses
- a) $\begin{pmatrix} 1 & 2 & 4 \\ -3 & 2 & 1 \end{pmatrix}$
 b) $\begin{pmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \end{pmatrix}$
 c) $\begin{pmatrix} 1 & 4 \\ 2 & -3 \\ 5 & 4 \end{pmatrix}$
 d) $\begin{pmatrix} 3 & 3 \\ 1 & 1 \\ 4 & 4 \end{pmatrix}$

Paragraph - II :

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ be a square matrix and C_1, C_2, C_3 be 3 column matrices satisfying $AC_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, AC_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ and $AC_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ of matrix B. If the matrix $C = \frac{1}{3}AB$ then

14. The value of sum of elements of B^{-1} is
 a) -1 b) 0 c) 4 d) 2
15. The ratio of trace of matrix A to the det of matrix B is
 a) 1 : 3 b) 2 : 3 c) 1 : 1 d) 3 : 1
16. The value of $\sin^{-1}|A| + \cos^{-1}|C|$ is (where $|A|$ is determinant of A)
 a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{4}$ d) 1

Matrix Matching Type Questions

17. Let $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$

Column-I

- A) A^{-1}
 B) $(adj A)^{-1}$
 C) $adj(adj A)$
 D) $adj(2A)$

Column-II

- p) $\begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$
 q) $2 \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$
 r) $\frac{1}{2} \begin{bmatrix} 1 + \cos 2x & -\sin 2x \\ \sin 2x & 1 + \cos 2x \end{bmatrix}$
 s) $\frac{1}{2} \begin{bmatrix} 1 + \cos 2x & \sin 2x \\ -\sin 2x & 1 + \cos 2x \end{bmatrix}$

18. **Column-I**

- A) If $|A|=2$, then $|2A^{-1}| =$ (where A is of order 3)
 B) If $|A|=1/8$, then $|adj(adj(2A))| =$ (where A is of order 3)
 C) If $(A+B)^2 = A^2 + B^2$, and $|A|=2$ then $|B| =$ (Where A and B are of odd order)
 D) $|A_{2 \times 2}| = 2, |B_{3 \times 3}| = 3$ and $|C_{4 \times 4}| = 4$, then $|ABC|$ is equal to

Column-II

- p) 1
 q) 4
 r) 24
 s) 0
 t) does not exist

Integer answer type questions

19. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 2 & 1 \\ 4 & 1 & 1 \\ 4 & 7 & 3 \end{bmatrix}$ then sum of squares of possible values of determinant of P is _____
20. If A & B are different matrices satisfying $A^3 = B^3$ and $A^2B = B^2A$ Find $|A^2 + B^2| =$ _____
21. If $A = \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$ and if $I + 2A + 3A^2 + \dots \infty = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ then numerical value of $|\alpha + \beta + \gamma + \delta|$ is _____
22. If A is a square matrix of order 3 such that $|A| = 2$, then $|(adj A^{-1})^{-1}|$ is _____
23. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ and $f(x)$ is defined as $f(x) = \det. (A^T A^{-1})$ then the value of $\underbrace{f(f(f(\dots f(x))))}_{n \text{ times}}$ is ($n \geq 2$) _____

EXERCISE-IVLinear equations**LEVEL-I (MAIN)**Single answer type questions

1. The rank of the matrix $\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{bmatrix}$ is _____
 1) 3 if $a = 6$ 2) 1 if $a = -6$ 3) 3 if $a = 2$ 4) 2 if $a = -6$
2. If the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & \alpha \end{bmatrix}$ is of rank 3, then $\alpha =$ _____
 1) -5 2) 5 3) 4 4) -4
3. If $c < 1$ and the system of equations $x + y - 1 = 0$, $2x - y - c = 0$, and $-bx + 3by - c = 0$ is consistent, then the possible real values of b are _____
 1) $b \in \left(-3, \frac{3}{4}\right)$ 2) $b \in \left(\frac{-3}{2}, 4\right)$ 3) $b \in \left(\frac{-3}{4}, 3\right)$ 4) $b \in \left(\frac{-3}{2}, \frac{3}{4}\right)$

4. Consider the system of equation $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ and $(a_1, b_1, a_2, b_2, c_1, c_2)$ are not all zero's) $\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \Delta_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$ then

Statement-I : If $\Delta = 0$ then the system is always consistent

Statement-II : If $\Delta \neq 0$ then the system is consistent

- 1) Both are true
2) Both are false
3) I is True, II is false
4) I is false, II is true
5. For the equations $x + 2y + 3z = 1, 2x + y + 3z = 2$ and $5x + 5y + 9z = 4$,
1) there is only one solution
2) there exists infinitely many solutions
3) there is no solution
4) there exists two solutions
6. The system of equations $x - y + 3z = 4, x + z = 2, x + y - z = 0$ has
1) a unique solution
2) finitely many solutions
3) infinitely many solutions
4) Two solutions
7. The system $x + 4y - 2z = 3, 3x + y + 5z = 7$ and $2x + 3y + z = 5$ has
1) infinite number of solutions
2) unique solution
3) trivial solution
4) no solution
8. Consider the system of linear equations
 $x_1 + 2x_2 + x_3 = 3; 2x_1 + 3x_2 + x_3 = 3; 3x_1 + 5x_2 + 2x_3 = 1$ The system has
1) no solution.
2) infinite number of solutions
3) exactly three solutions
4) a unique solution

9. Consider the system of equations $x - 2y + 3z = -1, x - 3y + 4z = 1$ and $-x + y - 2z = k$

Statement - I : The system of equations has no solution for $k \neq 3$

Statement - II : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$ for $k \neq 3$

- 1) Statement - I is true, Statement - II is true, Statement-II is a correct explanation for Statement - I
2) Statement - I is true, Statement - II is true, Statement - II is not a correct explanation for Statement - I
3) Statement - I is true, Statement - II false
4) Statement - I is false, Statement - II true
10. The system of equations $\alpha x + y + z = \alpha - 1; x + \alpha y + z = \alpha - 1; x + y + \alpha z = \alpha - 1$ has no solution if α is
1) -2
2) either -2 or 1
3) not -2
4) 1

11. Let a, b and c be any real numbers. Suppose that there are real numbers x, y and z not all zero such that $x = cy + bz, y = az + cx$, and $z = bx + ay$. Then $a^2 + b^2 + c^2 + 2abc$ is equal to
 1) 2 2) -1 3) 0 4) 1
12. The number of values of k for which the linear equations;
 $4x + ky + 2z = 0; kx + 4y + z = 0; 2x + 2y + z = 0$ Possesses a non-zero solution is
 1) zero 2) 3 3) 2 4) 1
13. The system of linear equations $x + \lambda y - z = 0, \lambda x - y - z = 0, x + y - \lambda z = 0$ has a non-trivial solution for
 1) Exactly one value of λ 2) Exactly two values of λ
 3) Exactly three values of λ 4) Infinitely many values of λ
14. If the system of linear equations $x + 2ay + az = 0; x + 3by + bz = 0; x + 4cy + cz = 0$ has a non-zero solution, then a, b and c
 1) are in A.P. 2) are in G.P.
 3) are in H.P. 4) satisfy $a + 2b + 3c = 0$.
15. Let the homogeneous system of linear equations $px + y + z = 0, x + qy + z = 0$ and $x + y + rz = 0$, where $p, q, r \neq 1$, have a non-zero solution, then the value of $\frac{1}{1-p} + \frac{1}{1-q} + \frac{1}{1-r}$ is
 1) -1 2) 0 3) 2 4) 1
16. The set of all values of λ for which the system of linear equations $2x_1 - 2x_2 + x_3 = \lambda x_1;$
 $2x_1 - 3x_2 + 2x_3 = \lambda x_2, -x_1 + 2x_2 = \lambda x_3$ has a non-trivial solution,
 1) is an empty set 2) is a singleton set
 3) contains two elements 4) contains more than two elements
17. The number of values of k , for which the system of equations: $(k+1)x + 8y = 4k;$
 $kx + (k+3)y = 3k - 1$ has no solution, is
 1) infinite 2) 1 3) 2 4) 3

LEVEL-II (ADVANCED)

Single answer type questions

1. The number of value of k for which the system of equations $(k+1)x + 8y = 4k$ and $kx + (k+3)y = 3k - 1$ has infinitely many solutions is
 a) 0 b) 1 c) 2 d) Infinite

2. Let a, b, c be the real numbers. Then following system of equation in x, y and z .

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1, \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, -\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \text{ has}$$

- a) no solution
b) unique solution
c) infinitely many solutions
d) finitely many solutions
3. Give $2x - y + 2z = 2, x - 2y - z = -4, x + y + \lambda z = 4$, then the value of λ such that the given system of equation has no solution is
- a) -3
b) 1
c) 0
d) 3
4. If the system of equations $x - ky - z = 0, kx - y - z = 0, x + y - z = 0$ has a nonzero solution, then the possible values of k are
- a) -1, 2
b) 1, 2
c) 0, 1
d) -1, 1
5. If the system of equations $2x + 3ky + (3k + 4)z = 0, x + (k + 4)y + (4k + 2)z = 0, x + 2(k + 4)y + (3k + 4)z = 0$ has non trivial solution then $k =$
- a) -8 or $\frac{1}{2}$
b) 8 or $-\frac{1}{2}$
c) -4 or $\frac{1}{2}$
d) 4 or $-\frac{1}{2}$

More than one correct answer type questions

6. Let $\alpha, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations $\alpha x + 2y = \lambda, 3x - 2y = \mu$. Which of the following statement(s) is correct?
- a) If $\alpha = -3$, then the system has infinitely many solutions for all values of λ and μ
b) If $\alpha \neq -3$, then the system has a unique solution for all values of λ and μ
c) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $\alpha = -3$
d) If $\lambda = \mu \neq 0$, then the system has no solution for $\alpha = -3$
7. The system of equations $-2x + y + z = a; x - 2y + z = b; x + y - 2z = c$ has
- a) no solution if $a + b + c \neq 0$
b) unique solution if $a + b + c = 0$
c) infinite number of solutions if $a + b + c = 0$
d) no solution if $a + b + c = 0$

Linked comprehension type questions

Passage- I :

Given that the system of equations $x = cy + bz, y = az + cx, z = bx + ay$ has non zero solutions and at least one of the a, b, c is a proper fraction.

8. $a^2 + b^2 + c^2$ is
- a) > 2
b) > 3
c) < 3
d) < 2
9. abc is
- a) > -1
b) > 1
c) < 2
d) < 3

MATRICES SYNOPOSIS & LS

(Objective)

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OBJECTIVE MATHEMATICS IA PART-1

MATRICES

10. System has solution such that

a) $x:y:z = (1-2a^2):(1-2b^2):(1-2c^2)$

b) $x:y:z = \frac{1}{1-2a^2}:\frac{1}{1-2b^2}:\frac{1}{1-2c^2}$

c) $x:y:z = \frac{a}{1-a^2}:\frac{b}{1-b^2}:\frac{c}{1-c^2}$

d) $x:y:z = \sqrt{1-a^2}:\sqrt{1-b^2}:\sqrt{1-c^2}$

Integer answer type questions

11. For what value of k does the following system of equations possess a non-trivial solution over the set of rationals $x+y-2z=0$, $2x-3y+z=0$, and $x-5y+4z=k$. Find all the solutions.
12. The set of equations $2x-3y=4$, $7x-2y=2$, $9x-5y=[4a]$ may be consistent, then $[a]$ is (where $[.]$ denotes G.I.F)
13. The system $ax+y+z=a$, $x+by+z=b$ and $x+y+cz=c$ has no solution if a,b,c are all different from the number -----
14. The number of all possible value of θ , where $0 < \theta < \pi$, for which the system of equations $(y+z)\cos 3\theta = xyz \sin 3\theta$, $x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$, $(xyz) \sin 3\theta = (y+2z)\cos 3\theta + y \sin 3\theta$ have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is
15. The value of θ for which the system $(\sin 3\theta)x - y + z = 0$, $(\cos 2\theta)x + 4y + 3z = 0$, $2x + 7y + 7z = 0$ has non-trivial solution is $\frac{5\pi}{\lambda}$, where λ is a positive integer greater than 1 then λ must be equal to

EXERCISE-V

Miscellaneous

LEVEL-I (MAIN)

Single answer type questions

1. If $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ then $\lim_{n \rightarrow \infty} \frac{1}{n} |A^n| =$

1) 1

2) 0

3) A

4) $\frac{1}{n} A$

2. Let $\omega = \frac{1}{2} + i\frac{\sqrt{3}}{2}$. Then the value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}$ is

1) 3ω

2) $3\omega(\omega-1)$

3) $3\omega^2$

4) $3\omega(1-\omega)$

3. In a ΔABC , the value of $\begin{vmatrix} e^{-i(2A)} & e^{iC} & e^{iB} \\ e^{iC} & e^{-i(2B)} & e^{iA} \\ e^{iB} & e^{iA} & e^{-i(2C)} \end{vmatrix}$ is (where $i = \sqrt{-1}$)

1) -2

2) -4

3) -3

4) -1

4. If $f(x) = ax^2 + bx + c$, $a, b, c \in R$ and equation $f(x) - x = 0$ has imaginary roots α and β and γ, δ

be the roots of $f\{f(x)\} - x = 0$ then $\begin{vmatrix} 2 & \alpha & \delta \\ \beta & 0 & \alpha \\ \gamma & \beta & 1 \end{vmatrix}$ is

- 1) 0
2) purely real
3) purely imaginary
4) non real complex numbers

5. If $g(x) = \begin{vmatrix} a^{-x} & e^{x \log_e a} & x^2 \\ a^{-3x} & e^{3x \log_e a} & x^4 \\ a^{-5x} & e^{5x \log_e a} & 1 \end{vmatrix}$ then

- 1) graphs of $g(x)$ is symmetrical about origin
2) graphs of $g(x)$ is symmetrical about y-axis
3) $\left. \frac{d^4 g(x)}{dx^4} \right|_{x=0} \neq 0$
4) $f(x) = g(x) \log \left(\frac{a-x}{a+x} \right)$ is an odd function

6. If $\Delta(x) = \begin{vmatrix} \alpha+x & \theta+x & \lambda+x \\ \beta+x & \phi+x & \mu+x \\ \gamma+x & \psi+x & v+x \end{vmatrix}$ then find the value of $\Delta'(x)$

- 1) $\alpha + \beta + \gamma$
2) 0
3) $2\alpha + \beta$
4) 3β

7. If $y = \sin mx$, then the value of the determinant $\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix}$, where $y_n = \frac{d^n y}{dx^n}$, is

- 1) m^3
2) m^2
3) m^3
4) 0

8. If $f'(x) = \begin{vmatrix} mx & mx-p & mx+p \\ n & n+p & n-p \\ mx+2n & mx+2n+p & mx+2n-p \end{vmatrix}$, then $y = f(x)$ represents

- 1) a straight line parallel to x-axis
2) a straight line parallel to y-axis
3) parabola
4) a straight line with negative slope

9. Let $f(x) = \begin{vmatrix} 2 \cos^2 x & \sin 2x & -\sin x \\ \sin 2x & 2 \sin^2 x & \cos x \\ \sin x & -\cos x & 0 \end{vmatrix}$. Then the value of $\int_0^{\pi/2} [f(x) + f'(x)] dx$ is

- 1) π
2) $\pi/2$
3) 2π
4) $3\pi/2$

10. The value of $\sum_{r=2}^n (-2)^r \begin{vmatrix} {}^{n-2}C_{r-2} & {}^{n-2}C_{r-1} & {}^{n-2}C_r \\ -3 & 1 & 1 \\ 2 & -1 & 0 \end{vmatrix} (n > 2)$ is

1) $2n-1+(-1)^n$

2) $2n+1+(-1)^{n-1}$

3) $2n-3+(-1)^n$

4) $2n-2+(-1)^n$

LEVEL-II (ADVANCED)

More than one correct answer type questions

1. Let a, b, c are non-zero real number and $\Delta = \begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix}$

where (ω) is cube root of unity, then Δ vanishes iff

a) $\frac{1}{a} + \frac{1}{b\omega} + \frac{1}{c\omega^2} = 0$

b) $\frac{1}{a} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0$

c) $\frac{1}{a\omega} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0$

d) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

2. If $f(x) = \begin{vmatrix} 3 & 3x & 3x^2 + 2a^2 \\ 3x & 3x^2 + 2a^2 & 3x^3 + 4a^2x \\ 3x^2 + 2a^2 & 3x^3 + 6a^2x & 3x^4 + 8a^2x^2 + 2a^4 \end{vmatrix}$ then

a) $f'(x) = 0$

b) $y = f(x)$ is a straight line parallel to x-axis

c) $\int_0^2 f(x) dx = 32a^4$

d) $f(x) = -16a$

3. Let $[A_k]_{n \times n}$ be a square matrix of order $n \times n$, such that $a_{ij} = \begin{cases} 0, i \neq j \\ \frac{1}{k+i}, i = j \end{cases}$ and $[B_k]_{n \times n}$ is its inverse matrix, then which is/are true?

a) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{k=1}^m \text{trace}(B_k)_{n \times n}}{m^3} \right) = \frac{1}{6}$

b) $\sum_{k=1}^{10} \text{trace}(B_k)_{n \times n} = 320$

c) $\lim_{m \rightarrow \infty} \left(\frac{\sum_{k=1}^m \text{trace}(B_k)_{n \times n}}{m^3} \right) = \frac{1}{3}$

d) $\sum_{k=1}^{10} \text{trace}(B_k)_{n \times n} = 330$

4. If A is symmetric and B is skew symmetric and $A+B$ is non singular and also $C = (A+B)^{-1}(A-B)$ then

a) $C^T(A+B)C = A+B$

b) $C^T(A-B)C = A-B$

c) $C^TAC = A$

d) $C^TAC = O$

5. If $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$, then $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I = lA^2 + mA + nI$, then

a) $3l - m + 2n = 4$

b) $2l + 3m - n = 4$

c) $l + m + n = 4$

d) $2l - m - n = 3$

Linked comprehension type questions

Passage - I :

Suppose $f(x)$ is a function satisfying the following conditions :

i) $f(0) = 2, f(1) = 1,$

ii) f has a minimum value at $x = 5/2$

iii) For all $x, f'(x) = \begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 2(ax+b) & 2ax+2b+1 & 2ax+b \end{vmatrix}$

6. The value of $f(2)$ is

a) $1/4$

b) $1/2$

c) -1

d) 3

7. $f(x)=0$ has

a) both roots positive

b) both roots negative

c) roots of opposite sign

d) imaginary roots

8. Range of $f(x)$ is

a) $[7/16, \infty)$

b) $(-\infty, 15/16]$

c) $[3/4, \infty)$

d) $(\frac{15}{16}, \infty)$

Passage - II :

A Pythagorean triplet is triplet of positive integers (a,b,c) such that $a^2 + b^2 = c^2$. Define matrices

P, Q, R by $P = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}, Q = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -2 \\ 2 & 2 & 3 \end{bmatrix}$ and $R = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$

9. If we write Pythagorean triplet (a,b,c) in matrix form as $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ then which of the following matrix product is not Pythagorean triplet?

a) $Q \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$

b) $P \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$

c) $R \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$

d) $\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$

15. Let α, β ($\alpha < \beta$) be the solutions of $\sum_{n=1}^6 \frac{1}{\sin\left(\theta + (n-1)\frac{\pi}{4}\right)\sin\left(\theta + \frac{n\pi}{4}\right)} = 4\sqrt{2}$ and determinant value

of A where $A = \begin{pmatrix} \sin \beta & \sin \alpha \\ \sin \alpha & \sin \beta \end{pmatrix}$ is k then $8k^2 =$

16. Let ω be the complex number $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex number z

satisfying $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$ is equal to

17. If $A = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$ and $A^8 = KA - \ell I$ then $K - \ell =$

18. Consider three matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ then the absolute value of

$\det(A) + \det\left(\frac{ABC}{2}\right) + \det\left(\frac{A(BC)^2}{4}\right) + \det\left(\frac{A(BC)^3}{8}\right) + \dots \infty$ is

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL - I

- 1) 4 2) 4 3) 2 4) 3 5) 3 6) 2 7) 1 8) 1
9) 2 10) 1 11) 2 12) 2 13) 3 14) 4 15) 4 16) 1
17) 3 18) 2 19) 2 20) 2 21) 4 22) 3 23) 4 24) 4
25) 1 26) 4 27) 1 28) 1 29) 1 30) 4 31) 3

LEVEL - II

- 1) b 2) d 3) d 4) b 5) d 6) ab 7) bcd 8) cd
9) d 10) a 11) b 12) c 13) c 14) c
15) A-s; B-p; C-q; D-r 16) 4 17) 3 18) 7 19) 5 20) 6

EXERCISE-II

LEVEL - I

- 1) 1 2) 2 3) 1 4) 1 5) 4 6) 2 7) 1 8) 1
9) 2 10) 3 11) 3 12) 1 13) 4 14) 2 15) 3 16) 3
17) 4 18) 1 19) 4 20) 4 21) 3 22) 2 23) 3 24) 4
25) 2 26) 3 27) 3 28) 3 29) 2 30) 2

LEVEL - II

- 1) a 2) c 3) b 4) b 5) a 6) a 7) b 8) b
 9) a 10) a 11) b 12) b 13) a 14) b 15) cd 16) bc
 17) bc 18) ac 19) abc 20) abcd 21) a 22) d 23) a 24) a
 25) b 26) c 27) d 28) c 29) c 30) c 31) d 32) b
 33) A-r; B-s; C-p; D-q 34) 2 35) 0 36) 2 37) 7 38) 5
 39) 7 40) 6 41) 0 42) 4 43) 2

EXERCISE-III**LEVEL - I**

- 1) 1 2) 1 3) 1 4) 4 5) 1 6) 4 7) 2 8) 2
 9) 1 10) 3 11) 2 12) 4 13) 4 14) 3 15) 4 16) 2
 17) 2 18) 1 19) 3 20) 1 21) 2 22) 3 23) 2 24) 4
 25) 1 26) 1 27) 3 28) 2 29) 1

LEVEL - II

- 1) a 2) c 3) ac 4) abcd 5) abc 6) bcd 7) abc 8) ad
 9) abd 10) ab 11) c 12) a 13) c 14) b 15) c 16) a
 17) A-r; B-s; C-p, D-q 18) A - q; B - p; C - s; D - t 19) 8 20) 0
 21) 2 22) 4 23) 1

EXERCISE-IV**LEVEL - I**

- 1) 2 2) 2 3) 3 4) 4 5) 1 6) 3 7) 4 8) 1
 9) 1 10) 1 11) 4 12) 3 13) 3 14) 3 15) 4 16) 3
 17) 2

LEVEL - II

- 1) b 2) b 3) d 4) d 5) a 6) bcd 7) ac 8) c
 9) a 10) d 11) 0 12) 1 13) 1 14) 3 15) 6

EXERCISE-V**LEVEL - I**

- 1) 2 2) 2 3) 2 4) 2 5) 1 6) 2 7) 4 8) 1
 9) 1 10) 1

LEVEL - II

- 1) abd 2) ab 3) ad 4) acd 5) ab 6) b 7) d 8) a
 9) b 10) d 11) c 12) A-r; B-r; C-q; D-s 13) 4 14) 1
 15) 6 16) 1 17) 1 18) 2